

Teja Naidu Chintha

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PROFESSIONAL SUMMARY

Data scientist with 4+ years building, validating, and monitoring machine learning models for fraud and credit risk in regulated financial services under SR 11-7 model risk governance. Hands-on across the full model lifecycle - developing champion-challenger XGBoost, LSTM, and deep learning models, then independently validating them through conceptual soundness review, backtesting. Proven at quantifying model risk, and presenting findings to risk, compliance, and executive stakeholders.

TECHNICAL SKILLS

Programming & Statistical: Python (Advanced), R, Scala, SQL, PySpark/Spark, Conda, REST APIs, Git

Model Risk & Validation: Model Validation, SR 11-7 Model Risk Governance, Conceptual Soundness, Backtesting, Outcomes Analysis, Champion-Challenger Benchmarking, Performance Monitoring, Annual Model Review, Model Governance Framework, Model Documentation

Statistical & ML Methods: Classification, Clustering, Regression, Anomaly Detection, Time Series Forecasting, Sentiment Analysis, A/B Testing, Statistical Inference (ANOVA), Confusion Matrix / ROC-AUC, KS / PSI

Models & Deep Learning: XGBoost, LightGBM, Gradient Boosting, Random Forest, Logistic/Linear Regression, LSTM/GRU, CNN, Autoencoders, Transformers (BERT/BioBERT), LLMs

Financial Crimes & Risk: Fraud Detection, Credit Risk Modeling, Transaction Monitoring, BSA/AML Analytics, Behavioral & Velocity Features

Data Engineering & MLOps: Apache Spark, Kafka, Airflow, Databricks, Delta Lake, ETL/ELT, MLflow, CI/CD

Cloud: AWS (SageMaker, Lambda, S3, EC2, Glue), Azure (Azure ML, Data Lake), Docker

Databases & Tools: PostgreSQL, MySQL, MongoDB, Snowflake, Alteryx, Jupyter, Vector Databases (FAISS, ChromaDB, Pinecone)

Visualization: Power BI, Tableau, Streamlit, Plotly

Domain: Model Risk Management, Financial Crimes, Fraud & AML, Regulatory & Responsible AI, Agentic AI Systems

PROFESSIONAL EXPERIENCE

U.S. Bank Oct 2024 – Present

Data Scientist, Fraud & Risk Analytics -- Irving, TX

- Led development of an end-to-end champion-challenger fraud detection system, replacing a legacy XGBoost champion with an LSTM challenger for card-not-present (CNP) fraud - achieving an 8% relative improvement in PR-AUC (0.52 to 0.56), a 12% lift in fraud detection rate, and approximately \$2.5M in annual fraud loss reduction.
- Architected and deployed a real-time Kafka/Lambda/DynamoDB scoring pipeline for CNP fraud models, engineering behavioral and velocity features over streaming transaction data and managing CI/CD versioning and rollback to ensure production stability.
- Built LLM-powered investigation agents to mine textual case data, automate financial-crime decision support, and reduce manual review effort - with full explainability and audit logging integrated to meet regulated-workflow compliance requirements.
- Designed model evaluation frameworks measuring discriminatory power and accuracy using confusion matrices, ROC/AUC, KS, and PSI; authored validation findings and presented quantified model risks and business impact to risk, compliance, and executive stakeholders.
- Established ongoing model performance monitoring and observability pipelines (MLflow, AWS CloudWatch) tracking score drift, PSI, and alert rates to support annual model review and recalibration cycles, reducing model degradation detection lag.
- Executed independent model validation across classification, clustering, anomaly detection, sentiment analysis, and time-series models under SR 11-7 model risk governance - covering conceptual soundness review, data integrity assessment, model controls, and complete lifecycle documentation aligned with regulatory standards.
- Partnered with a cross-functional team of data scientists, software engineers, and product managers to translate business risk requirements into model specifications, coordinate staged deployments, and deliver validation findings to compliance and executive audiences.
- Mentored junior analysts on model validation best practices, champion-challenger experimental design, and SR 11-7 documentation standards - fostering a culture of model risk awareness and analytical rigor within the fraud analytics team.

Heartland Community Network Jul 2023 – Sep 2024

Data Scientist -- Remote, Bloomington, IN

- Built time-series forecasting models (Prophet, LSTM) projecting patient visit volumes for IU Health clinic networks - achieving a 14% reduction in mean absolute error versus the prior baseline, enabling more accurate staffing and resource allocation decisions.
- Developed BioBERT-based NLP classification pipeline over clinical notes to identify high-risk patient cohorts, delivering structured risk flags that reduced manual chart review time by approximately 30% across partner health system teams.
- Designed validation and benchmarking routines comparing competing model variants - documenting assumptions, limitations, and performance metrics (accuracy, stability, error analysis) in stakeholder-ready reports reviewed by clinical and operational leads.
- Deployed reproducible analytics workflows on Azure (Data Factory, Delta Lake, Azure ML) with version control and controlled releases, maintaining full audit trails and data lineage documentation aligned with healthcare data governance requirements.

Cognizant (Client: TD Bank) Jan 2021 – Jul 2022

Data Analyst -- Remote

- Designed and maintained transaction-monitoring SQL pipelines processing millions of financial records daily, building transformation logic that surfaced anomalous patterns and risk indicators used by TD Bank's financial intelligence and AML investigation teams.
- Built Power BI and Tableau dashboards tracking financial-crime KPIs and model output for compliance stakeholders, replacing manual reporting processes and cutting dashboard refresh time from weekly to near-real-time.
- Assessed data integrity across large-volume numeric financial datasets - identifying inconsistencies, mapping upstream data-quality issues, and implementing validation checks that improved the reliability of AML analytic outputs.
- Supported BSA/AML reporting and financial-intelligence workflows in alignment with regulatory requirements, ensuring SQL transformation logic and data outputs met audit and compliance documentation standards.
- Collaborated with cross-functional stakeholders across data engineering, compliance, and risk teams to translate business reporting requirements into scalable SQL and BI solutions, presenting findings and data insights to senior compliance managers.

PROJECTS

Ledger - Agentic Data Analyst Platform ledger-analyst.streamlit.app

- Designed and deployed a multi-agent agentic data science pipeline that autonomously ingests user-uploaded datasets, selects analytical approaches, generates Python code, executes it in a sandboxed environment, and returns charts and narrative insights - end-to-end without manual intervention.
- Architected a LangGraph-based orchestration layer coordinating specialized agents (data profiling, feature analysis, visualization, narrative generation) with shared state management, enabling complex multi-step analyses through a single natural-language prompt.
- Live at ledger-analyst.streamlit.app - built as a portfolio demonstration of production-grade agentic AI system design, with structured error handling, agent tracing, and incremental result streaming for real-time user feedback.

Python · LangGraph · Streamlit · Pandas · OpenAI API · Plotly

Regression Model Selection - Yield Prediction Indiana University

- Evaluated and compared regression approaches (Lasso, Ridge, Gradient Boosting) for predictive accuracy and stability, documenting assumptions and trade-offs through validation-style analysis; deployed a Streamlit app for live inference and result inspection.

Python · Scikit-learn · Gradient Boosting · Model Selection · Streamlit

EDUCATION

Indiana University Bloomington Aug 2022 – May 2024

M.S. in Data Science - Bloomington, Indiana

SRM University Jun 2017 – Jun 2021

B.Tech in Computer Science - Chennai, India

CERTIFICATIONS

IBM Data Science Professional Certificate IBM, 2026

Google AI Professional Certificate Google, 2026

AWS Certified Solutions Architect – Associate Amazon Web Services, 2024

Alteryx Designer Advanced Alteryx, 2025

Alteryx Designer Core Alteryx, 2025